

NOBEL DHAR

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SUMMARY

Researcher in efficient **LLM inference** on memory-constrained hardware (PhD, May 2027). One problem drives this work: large language models are outgrowing the memory of the devices meant to run them. My research established that **memory, not compute, is the binding constraint** on edge devices, then dismantled it — **creating activation sparsity in models that have none**, making that sparsity **predictable at scale** by clustering activation patterns into centroids, and **fetching only the weights the next token needs**. The result, **NeuroPrefetcher**, runs models larger than device memory **7.9–12.0× faster than llama.cpp**. Current work extends the same principle to **mixture-of-experts routing and the KV cache**. 11 publications, **166 citations**, **h-index 6**; cited by researchers at Meta, Google, NVIDIA, Stanford, and MIT. Co-inventor on a U.S. patent application.

EDUCATION

Kennesaw State University PhD, Computer Science — GPA 3.71/4.0	Marietta, GA Aug 2023 – May 2027 (expected)
– Dissertation: <i>Bridging the Model–Memory Gap: Intelligent Storage I/O for Scalable Neural Network Inference on Resource-Constrained Platforms</i> (advisor: Dr. Kun Suo; proposal defended May 2026)	
Shahjalal University of Science and Technology BS, Computer Science	Sylhet, Bangladesh 2016 – 2021

RESEARCH EXPERIENCE

Graduate Research Assistant Kennesaw State University	Marietta, GA Aug 2023 – Present
The Memory Wall — first cross-tier empirical study of on-device LLMs [data & code] ACMSE 2024 · 80 citations	
– LLaMA-2 7B INT4 from Raspberry Pi 4B to Jetson AGX Orin : 0.11 → 4.49 tok/s (8 GB Pi → Orin), a 40.8× gap; memory, not compute, is the binding constraint	
– Cited by groups at Meta, Google + UC Berkeley, NVIDIA, Stanford, MIT-affiliated, and UCSD — the constraint the next three papers attack	
Enforced Activation Sparsity — created where SwiGLU models have none [code]	IEEE IPCCC 2024
– ~50% FFN activation sparsity without retraining on Llama-3-8B, Mistral-7B, Phi-3, and OPT, with negligible perplexity loss — the sparsity NeuroPrefetcher later exploits	
Activation-Aware Clustering — sparsity prediction made feasible [code]	Euro-Par 2025
– Reformulated per-neuron prediction as centroid-bank classification, cutting per-token prediction cost $\approx 760,000\times$	
– Custom sparsity-aware clustering (active-neuron-only distance, balanced assignment, percentile centroid update): 79.3% gate-projection precision vs. 70.6% BMF and 69.6% BRB-KMeans over 10.6M activation patterns	
NeuroPrefetcher — runs LLMs larger than device memory from NVMe [code]	ICPP 2026
– 7.9–12.0× over llama.cpp (3.22 vs. 0.34 tok/s) on Jetson AGX Orin; fastest of the only 4 of 20 production systems that generate — vLLM and TensorRT-LLM fail before output	
– One GPU-resident activation predictor (2.86% of base LLM parameters) plans all 31 sparse layers' I/O after layer 0; ~99% correct centroids, 95–96% dense HellaSwag retained	
– Trained the sparsity predictor from scratch : DDP/NCCL on 8× A100-80GB, AdamW, batch 2,048, 1,000 epochs	
– Delta prefetching reuses 82–85% of active neurons per token; io.uring + O_DIRECT (QD 256, 7.4 GB/s), pinned staging, fused CUDA scatter kernel	
– Page faults 11,453/s → 0.21/s , CPU iowait 77% → 4%, GPU utilization 3% → 30% vs. OS demand paging	
TIDE — elastic memory for mixture-of-experts inference (current work)	in prep.
– Routing study of Qwen3-30B-A3B (128 experts/layer, top-8) on Jetson AGX Thor: a frequency-ranked resident pool covers 99.5% of decode-time expert accesses	
– mmap/madvise pressure governor sheds 8.9 GiB of expert weights in 0.4 s mid-generation; output byte-identical, within 10% of stock llama.cpp	
– KV-cache balloon commits memory only as tokens are written: 40,960-token context inside a 16 GiB budget where full residency is OOM-killed — the same memory-first principle on MoE experts and the KV cache	

Dhar, N., Islam, M.R., Zhang, X., Sun, G., Islam, S., Deng, B., Suo, K. “NeuroPrefetcher: Storage-Aware Sparse LLM Inference via Delta Prefetching.” [International Conference on Parallel Processing \(ICPP\) 2026](#) [arXiv] [code]

Dhar, N., Deng, B., Islam, M.R., Zhang, X., Nasif, K.F.A., Suo, K. “A Sparsity Predicting Approach for Large Language Models via Activation Pattern Clustering.” [International European Conference on Parallel and Distributed Computing \(Euro-Par\) 2025](#), Springer LNCS 15900 — 1 citation [code]

Dhar, N., Deng, B., Islam, M.R., Nasif, K.F.A., Zhao, L., Suo, K. “Activation Sparsity Opportunities for Compressing General Large Language Models.” [IEEE International Performance, Computing, and Communications Conference \(IPCCC\) 2024](#) — 6 citations [code]

Dhar, N., Deng, B., Lo, D., Wu, X., Zhao, L., Suo, K. “An Empirical Analysis and Resource Footprint Study of Deploying Large Language Models on Edge Devices.” [ACM Southeast Conference \(ACMSE\) 2024](#) — 80 citations [data & code]

Dhar, N., Saha, G., Bhattacharjee, P., Mallick, A., Islam, M.S. “Pointer over Attention: An Improved Bangla Text Summarization Approach Using Hybrid Pointer Generator Network.” [International Conference on Computer and Information Technology \(ICCIT\) 2021](#) — 7 citations

Islam, M.R., **Dhar**, N., Deng, B., Nguyen, T.N., He, S., Suo, K. “Characterizing and Understanding the Performance of Small Language Models on Edge Devices.” [IEEE International Performance, Computing, and Communications Conference \(IPCCC\) 2024](#) — 34 citations

Islam, M.R., Deng, B., **Dhar**, N., Nguyen, T.N., He, S., Shi, Y., Suo, K. “Characterizing and Understanding Energy Footprint and Efficiency of Small Language Models on Edges.” [IEEE International Conference on Mobile Ad-Hoc and Smart Systems \(MASS\) 2025](#) — 4 citations

Islam, M.R., Vu, L., **Dhar**, N., Deng, B., Suo, K. “Building a Resilient and Sustainable Grid: A Study of Challenges and Opportunities in AI for Smart Virtual Power Plants.” [ACM Southeast Conference \(ACMSE\) 2024](#) — 18 citations

Suo, K., Vu, L., Islam, M.R., **Dhar**, N., Nguyen, T.N., He, S., Wu, X. “A Systematic Investigation of Hardware and Software in Electric Vehicular Platform.” [ACM Southeast Conference \(ACMSE\) 2024](#) — 9 citations

Vu, L., Suo, K., Islam, M.R., **Dhar**, N., Nguyen, T.N., He, S., Shi, Y. “Living on the Electric Vehicle and Cloud Era: A Study of Cyber Vulnerabilities, Potential Impacts, and Possible Strategies.” [ACM Southeast Conference \(ACMSE\) 2024](#) — 6 citations

Nasif, K.F.A., Deng, B., Chen, L., Xie, Y., Teng, S., Li, J., Zhao, L., Alam, S.M.S., **Dhar**, N., Suo, K., Lo, D.C.T. “AccelMD: A Self-Adaptive AI-Enabled Framework for Accelerating Molecular Dynamics Simulations.” [IEEE International Conference on Tools with Artificial Intelligence \(ICTAI\) 2025](#) — 1 citation

PATENT

Co-inventor, “Systems and Methods for Using Edge Devices in Multi-Modal Vision Models,” U.S. provisional application No. 52494 (filed Apr 2025), assigned to InComm Payments — Go Studio × KSU industry–academic collaboration

INDUSTRY EXPERIENCE

Software Engineer <i>Bangladesh Japan Information Technology (BJIT)</i>	Dhaka, Bangladesh Aug 2021 – Jun 2023
– Shipped two AOSP system apps for PockeTalk, a commercial embedded translation device: device Settings, and an eSIM manager bridging web services to the SIM chip through the HAL – Built its translation app (speech, text, image) on five cloud inference providers — Google, Microsoft, Baidu, DeepL, Toshiba — since the device could not run models locally	
Associate Specialist, Software Development <i>ParallaxLogic Infotech</i>	Dhaka, Bangladesh Feb 2021 – Aug 2021
– Built consumer and merchant Android apps for ekShop, a Bangladesh government e-commerce platform — Kotlin, MVVM, REST APIs, dependency injection, SQLite, Firebase	

TECHNICAL SKILLS

LLM & ML: PyTorch, transformer internals (attention, KV cache, SwiGLU/GELU FFN, MoE expert routing), activation sparsity, quantization (INT4), structured pruning, distributed training (DDP, NCCL, FP16/BF16), HellaSwag/WikiText evaluation

GPU & Systems: CUDA C++ (fused kernels, vectorized loads, streams, pinned memory), Nsight profiling, bottleneck decomposition (page faults, iowait, GPU utilization) with sysstat (mpstat, iostat), tegrastats, nmon, and Linux /proc counters (majflt, MemAvailable, PSI, cgroup accounting), multi-GPU A100 clusters

Storage & OS: Linux io_uring, O_DIRECT, NVMe block I/O, page-fault analysis, mmap/madvise paging control, cgroup memory limits, custom binary formats

Edge & Embedded: NVIDIA Jetson AGX Orin/Thor (JetPack, unified memory), Raspberry Pi, AOSP/HAL, embedded Android

Languages: Python, C++, CUDA C++, Java, Kotlin, SQL

SERVICE & RECOGNITION

Peer reviewer, IEEE ICTAI 2025 | Sigma Xi (nominated 2026)